

Name.....

Roll No.....


ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.
(An ISO 9001:2015 Certified College)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B. TECH DEGREE FIRST INTERNAL TEST-OCTOBER 2023

Seventh Semester

CST 401 - ARTIFICIAL INTELLIGENCE

Time: 2 hours

Max. Marks: 50

PART - A Answer all questions (3 x 5 = 15 Marks)

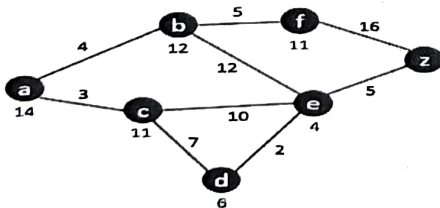
Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	Describe how intelligent agents interact with environment.	3	K2	CO1 & K2
2	Discuss the working of Utility based agent.	3	K2	CO1 & K2
3	Explain the components of a Well-defined problem and solution.	3	K2	CO2 & K2
4	How do you overcome the limitations of Depth limited search.	3	K1	CO2 & K2
5	Explain the major types of games with examples.	3	K2	CO3 & K3

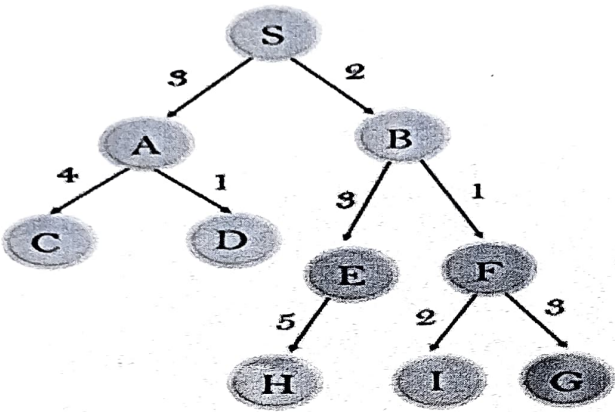
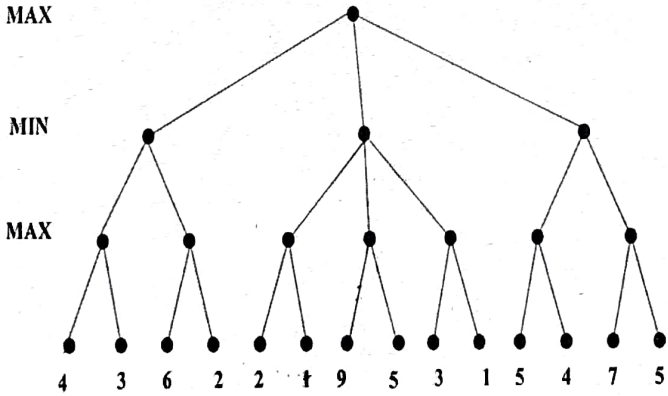
PART - B (Answer any 2 Questions)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
6	Describe the different types involved in task environment.	7	K2	CO1 & K2
7	What is an agent program? Illustrate the working of Goal-based and Model reflex agent programs.	7	K2	CO1 & K2
8	List and explain any 7 applications of Artificial Intelligence in recent times.	7	K2	CO1 & K2

PART - C (Answer any 2 Questions)

9	Find the optimal solution for the given graph using A* algorithm.	7	K2	CO2 & K2
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10	Explain the purpose of Heuristic function and Identify the heuristics for 8-puzzle problem.	7	K2	CO2 & K2
11	Consider the following tree with node S as start node and node G as Goal node. Search the tree, using the following algorithms a) Uniform Cost Search b) Depth First Search 	7	K2	CO2 & K2
PART - D(Mandatory)				
12	Consider the following game tree; Explain the working of alpha beta pruning with respective moves and steps. 	7	K3	CO3 & K3

Name.....

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**ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.***(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****B. TECH DEGREE SECOND INTERNAL TEST-DECEMBER 2023**

Seventh Semester

CST 401 & ARTIFICIAL INTELLIGENCE

Time: 2 hours

Max. Marks: 50

PART - A Answer all questions (3 x 5 = 15 Marks)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
1	With relevant diagram explain working of knowledge based agent.	3	K2	CO4 & K3
2	Find the Most General Unifier for $Q(a, g(x, a), f(y))$ and $Q(a, g(f(b), a), x)$.	3	K3	CO4 & K3
3	Define the terms overfitting and underfitting with examples.	3	K1	CO5 & K2
4	State and explain Ockham's razor principle.	3	K1	CO5 & K2
5	What are the components of a Constraint Satisfaction Problem? Illustrate with an example.	3	K2	CO3 & K3

PART - B (Answer any 2 Questions)

Qn. No.	Question	Marks	K level of Qn.	CO & KL
6	Convert the following sentences to First Order Logic a) Everyone who loves all animals is loved by someone. b) Anyone who kills an animal is loved by no one. c) Jack loves all animals d) Either Jack or Curiosity killed the cat, who is named Tuna.	7	K3	CO4 & K3
7	Convert the following sentence to CNF [(Food $\rightarrow$ Party) $\vee$ (Drinks $\rightarrow$ Party)] $\rightarrow$ [(Food $\wedge$ Drinks) $\rightarrow$ Party].	7	K3	CO4 & K3
8	Consider the statement given below. Explain the working of Forward chaining. "As per the law, it is a crime for an American to sell weapons to hostile nations. Country A, an enemy of America, has some missiles, and all the missiles were sold to it by Robert, who is an American citizen." Prove that "Robert is criminal."	7	K3	CO4 & K3

PART - C (Answer any 2 Questions)

9	Explain the different forms of learning in Machine Learning.	7	K2	CO5 & K2																																												
10	a) Define Regression with a suitable example.	2	K2	CO5 & K2																																												
	b) Explain the different types of linear regression.	5																																														
11	Consider the following dataset comprised of three input attributes (Age, Competition, Type) and one binary output(Class). Use the Decision tree learning algorithm to learn a decision tree for these data. Show the computations made to determine the attribute to split at each node. <table><tr><th>Age</th><th>Competition</th><th>Type</th><th>Class (Profit)</th></tr><tr><td>Old</td><td>Yes</td><td>Software</td><td>Down</td></tr><tr><td>Old</td><td>No</td><td>Software</td><td>Down</td></tr><tr><td>Old</td><td>No</td><td>Hardware</td><td>Down</td></tr><tr><td>Mid</td><td>Yes</td><td>Software</td><td>Down</td></tr><tr><td>Mid</td><td>Yes</td><td>Hardware</td><td>Down</td></tr><tr><td>Mid</td><td>No</td><td>Hardware</td><td>Up</td></tr><tr><td>Mid</td><td>No</td><td>Software</td><td>Up</td></tr><tr><td>New</td><td>Yes</td><td>Software</td><td>Up</td></tr><tr><td>New</td><td>No</td><td>Hardware</td><td>Up</td></tr><tr><td>New</td><td>No</td><td>Software</td><td>Up</td></tr></table>	Age	Competition	Type	Class (Profit)	Old	Yes	Software	Down	Old	No	Software	Down	Old	No	Hardware	Down	Mid	Yes	Software	Down	Mid	Yes	Hardware	Down	Mid	No	Hardware	Up	Mid	No	Software	Up	New	Yes	Software	Up	New	No	Hardware	Up	New	No	Software	Up	7	K2	CO5 & K2
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PART - D(Mandatory)

12	Solve the following crypt arithmetic problem by hand, using the strategy of backtracking with forward checking and the MRV & least-constraining-value heuristics. E A T + T H A T A P P L E	7	K3	CO3 & K3
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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 Scheme)

Course Code: CST401**Course Name: ARTIFICIAL INTELLIGENCE****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- | | | |
|------|---|-----|
| 1 ✓ | Define PEAS representation. Give the PEAS representation of a self-driving car. | (3) |
| 2 ✓ | Distinguish between episodic and sequential environment. | (3) |
| 3 | Describe the state space representation of 8-queens problem. | (3) |
| 4 | Discuss the infrastructure needed to solve a search problem. | (3) |
| 5 ✓ | Explain the properties of min-max algorithm. | (3) |
| 6 ✓ | Define Constraint Satisfaction Problem with an example. | (3) |
| 7 ✓ | Explain Modus Ponens and Modus Tollens with an example. | (3) |
| 8 ✓ | Discuss the drawbacks of propositional logic. | (3) |
| 9 ✓ | Explain cross validation in machine learning. | (3) |
| 10 ✓ | Distinguish between supervised and unsupervised learning. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

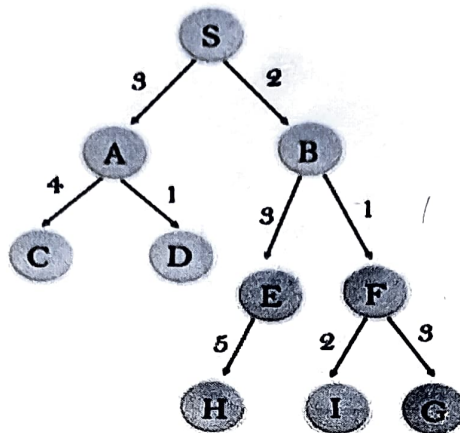
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|----|--|-----|
| 11 | a) Explain learning agent with a neat diagram. | (8) |
| | b) Discuss the milestones in the history of Artificial Intelligence. | (6) |

OR

- | | | |
|----|---|-----|
| 12 | a) ✓ Explain the structure of a model based reflex agent with a neat diagram. | (8) |
| | b) ✓ Describe the aspects of AI based on human centred approach. | (6) |

Module II

- | | | |
|------|---|------|
| 13 ✓ | Explain Best First Search algorithm. Perform BFS search in the problem given below to reach goal node G from source node S. | (14) |
|------|---|------|



The heuristic values of A, B, C, D, E, F, G, H, I, S, G are 12, 4, 7, 3, 8, 2, 4, 9, 13, and 0 respectively.

OR

- 14 a) Explain the working of Depth Limited Search with an example. (8)
 b) Discuss the optimality of A* algorithm. (6)

Module III

- 15 Solve the following crypt arithmetic problem using MRV and forward checking. (14)

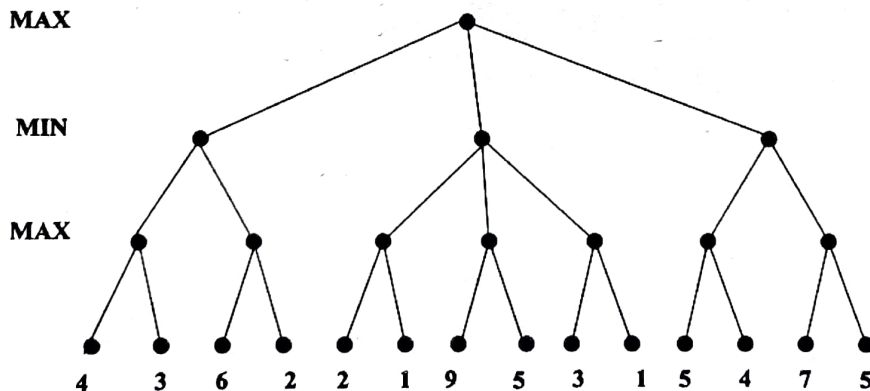
SEND +

MORE

MONEY

OR

- 16 Perform Alpha beta pruning in the two player game tree given below. (14)



Module IV

- 17 a) Explain the syntax and semantics of predicate logic. (10)
 b) Define entailment with an example. (4)

OR

- 18 a) Consider the following facts in a knowledge base (8)

1. Gita likes all kinds of food.
2. Mango and chapati are food.
3. Gita eats almond and is still alive.
4. Anything eaten by anyone and is still alive is food.

Prove "Gita likes Almond" using resolution technique.

- b) Distinguish between forward chaining and backward chaining. (6)

Module V

- 19 Explain linear regression. Use the following data to construct a linear regression model for the auto insurance premium as a function of driving experience. (14)

Driving experience (in years)	5	2	12	9	15	6	25	16
Auto insurance premium (\$)	64	87	50	71	44	56	42	60

OR

- 20 Explain the working of a decision tree. Use ID3 algorithm and find the best attribute to split at the root level of a decision tree. (14)

Age	Competition	Type	Class (profit)
Old	Yes	Software	Down
Old	No	Software	Down
Old	No	Hardware	Down
Mid	Yes	Software	Down
Mid	Yes	Hardware	Down
Mid	No	Hardware	Up
Mid	No	Software	Up
New	Yes	Software	Up
New	No	Hardware	Up
New	No	Software	Up
